MIL-S-43104B

27 February 1968

SUPERSEDING
MIL-S-43104A(GL)
3 September 1964

MILITARY SPECIFICATION

STRAPPING AND SEALING KITS, HAND-OPERATED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

★ 1.1 Scope.- This specification covers hand-operated, flat steel strapping and sealing kits.

★ 1.2 Classification.- Steel strapping and sealing kits shall be of the following types, as specified (see 6.2).

Type I	-	3/8-inch wide strapping.
Type II	-	1/2-inch wide strapping.
Type III	-	5/8-inch wide strapping.
Type IV	-	3/4-inch wide strapping.
Type V	-	1-1/4-inch wide strapping.
Type VI	-	2-inch wide heavy duty strapping.

★ 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film (Polyethylene Thin Gauge).
QQ-S-781	-	Steel Strapping, Flat.
GGG-C-835	-	Cutter, Steel Strapping.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-S-17743	-	Stretchers, Steel Strapping, Hand.
MIL-S-43180	-	Sealer, Steel Strapping, Hand, Nonpowered.

FSC 3540

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-731 - Quality of Wood Members for Containers and Pallets.

HANDBOOK

MILITARY

MIL-HDBK-7 - Lumber and Allied Products.

DRAWINGS

ARMY NATICK LABORATORIES

5-7-64 - Strapping and Sealing Kits, Hand-Operated; Details, Box, Kit (For components of Types I, II, III and IV Kits).
5-7-65 - Strapping and Sealing Kits, Hand-Operated; Details, Box, Kit (For components of Type V Kit).
5-7-66 - Strapping and Sealing Kits, Hand-Operated; Details, Box, Kit (For components of Type VI Kit).

(Miniature copies of Drawings 5-7-64 through 5-7-66, identified as figures 1 through 3, are attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.2), before production is commenced, a sample strapping and sealing kit conforming to this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Materials.- Materials shall conform to referenced specifications, drawings, and requirements specified hereinafter.

* 3.2.1 Lumber, solid, 3/4-inch thick.- The 3/4-inch thick solid lumber shall conform to wood specie groups I or II of MIL-STD-731; S1S2E or S4S (MIL-HDBK-7).

* 3.2.2 Hardware.- The hardware shall be commercially available, conforming to the type, size, material and finish specified on Drawings 5-7-64 through 5-7-66.

* 3.3 Components of kits.- The components of the type I through type VI kits shall be as listed in tables I through VI, respectively. The quantity required for one kit shall be as indicated in the applicable table.

* 3.3.1 Stretcher, hand-operated.- The hand-operated stretcher shall conform to type II, style 1 or type I, style 1 of MIL-S-17743. The stretcher shall be for use with the size strapping furnished in the particular kit.

3.3.2 Steel strapping, flat.- The flat steel strapping shall conform to type I, class B, grade 2 flat edges of QQ-S-781.

TABLE I.- Type I kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 3/8 inch by 0.020 inch strapping.	3.3.1
1	Sealer, steel strapping, hand; for 3/8 inch by 0.020 inch strapping.	3.3.6
2000	Seal, steel strapping, Interlake Steel Corp. No. 31 or equal.	
90 pounds minimum	Steel strapping, flat, 3/8 inch by 0.020 inch.	3.3.2
1	Box for kit components.	3.3.3

TABLE II.- Type II kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 1/2 inch by 0.020 inch flat steel strapping.	3.3.1
1	Sealer, steel strapping, hand; for 1/2 inch by 0.020 inch strapping.	3.3.6

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TABLE II.- Type II kit components (cont'd)

Quantity required	Components	Reference paragraph
2000	Seal, steel strapping, Interlake Steel Corp. No. 41 or equal.	
90 pounds minimum	Steel strapping, flat, 1/2 inch by 0.020 inch.	3.3.2
1	Box for kit components.	3.3.3

TABLE III.- Type III kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 5/8 inch by 0.020 inch strapping.	3.3.1
1	Sealer, steel strapping, hand; for 5/8 inch by 0.020 inch strapping.	3.3.6
2000	Seal, steel strapping, Interlake Steel Corp. No. 51 or equal.	
90 pounds minimum	Steel strapping, flat, 5/8-inch by 0.020 inch.	3.3.2
1	Box for kit components.	3.3.3

TABLE IV.- Type IV kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 3/4 inch by 0.020 inch strapping.	3.3.1
1	Sealer, steel strapping, hand; for 3/4-inch by 0.020 inch strapping.	3.3.6
2000	Seal, steel strapping, Interlake Steel Corp. No. 61 or equal.	
90 pounds minimum	Steel strapping, flat, 3/4 inch by 0.020 inch.	3.3.2
1	Box for kit components.	3.3.3

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TABLE V.- Type V kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 1-1/4 inch by 0.035 inch strapping.	3.3.4
1	Sealer, steel strapping, hand; for 1-1/4 inch by 0.035 inch strapping.	3.3.6
1	Cutter, steel strapping, heavy duty, forward cutting action.	3.3.5
100	Seal, steel strapping, Interlake Steel Corp. No. 1032 or equal.	
90 pounds minimum	Steel strapping, flat 1-1/4 inch by 0.035 inch.	3.3.2
1	Box for kit components.	3.3.3

TABLE VI.- Type VI kit components

Quantity required	Components	Reference paragraph
1	Stretcher, steel strapping, hand; for 2 inch by 0.050 inch strapping.	3.3.4
1	Sealer, steel strapping, hand; for 2 inch by 0.050 inch strapping.	3.3.6
1	Cutter, steel strapping, heavy duty, forward cutting action.	3.3.5
50	Seal, steel strapping, Interlake Steel Corp. No. 1642 or equal.	
90 pounds minimum	Steel strapping, flat, 2 inch by 0.050 inch.	3.3.2
1	Box for kit components.	3.3.3

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* 3.3.3 Box, kit components.- The kit component box shall conform to the material, design and details shown on the following drawings:

Drawing 5-7-64 - Type I through type IV kit boxes
Drawing 5-7-65 - Type V kit box
Drawing 5-7-66 - Type VI kit box

3.3.4 Stretcher, steel strapping, hand; for 1-1/4 inch and 2 inch heavy duty strapping.- The hand-operated steel strapping stretcher shall conform to type III, style 1, without cutoff of MIL-S-17743, and shall be for use with the size strapping furnished in the particular kit.

3.3.5 Cutter, steel strapping.- The steel strapping cutter shall conform to type II of GGG-C-835.

* 3.3.6 Sealer, steel strapping, hand, nonpowered.- The hand-operated, steel strapping sealer shall conform to MIL-S-43180 and shall be furnished in the type and size applicable to the size strapping and type of seal furnished in the particular kit.

* 3.4 Operating instructions.- Instructions covering the operation of the sealer and stretcher shall be furnished with each specified type.

3.5 Marking.- The marking requirements shall conform to MIL-STD-130 and comply with Drawings 5-7-64, 5-7-65, 5-7-66.

* 3.6 Workmanship.- The finished strapping and sealing kits shall be free from defects such as splits, sharp edges, slivers or projections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects specified in tables VII and VIII.

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* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.1.1 Visual and dimensional examination.- A visual and dimensional examination shall be made of the kit boxes to determine compliance with Drawings 5-7-64, 5-7-65 and 5-7-66, as applicable. Defects shall be as specified in table VII. A lot shall be all the kit boxes of one size offered for inspection at one time. The sample unit for this inspection shall be one box. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

TABLE VII.- Visual and dimensional examination for boxes

Examine	Defect
Markings (interior)	mitted; incorrect; illegible; of improper size or location.
Material	Screw, hasps, and sash cord loose or missing.
Finish	Not as specified.
Dimensions	Not as specified.

* 4.3.2 Inspection of the end product.- A lot shall be all the hand sealing kits of one type offered for inspection at one time. The sample unit for this inspection shall be one complete hand sealing kit.

* 4.3.2.1 Visual examination.- Visual examination shall be made of the kits for the defects set forth in table VIII. The inspection level shall be I and with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE VIII.- Visual examination for kits

Examine	Defect
Assembly	Any component missing. Operating instructions for stretch and sealer not furnished with each kit.
Marking	Omitted, illegible, incorrect, of improper size or location.

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* 4.3.3 Examination of preparation for delivery.-

* 4.3.3.1 Seals.- An examination shall be made to determine that packaging, packing and marking comply with the requirements of section 5 of this specification for seals. Defects shall be scored as specified in table IX. The sample unit shall be one box of seals. The lot shall be the number of boxes of seals of one size offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE IX.- Examination of packaging of seals

Examine	Defect
Seals	Wrong type. Quantity of seals more or less than specified. Bag missing.

* 4.3.3.2 Examination of preservation.- Examination of preservation shall be in accordance with MIL-P-116.

* 4.3.3.3 Examination of containers packing and marking.- An examination shall be made of kits to determine that packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table X. The sample unit shall be one kit fully prepared for delivery. The lot shall be the number of packed kits offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE X.- Examination of packing and marking

Examine	Defect
Markings (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Strapping or seals damaged. Operating instructions not properly attached.
Workmanship	Inadequate application of components such as loose strapping, inadequate stapling; bulging or distortion of containers. Components not properly placed.
Contents	More or less than required.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.-

* 5.1.1.1 Sealers, stretchers and cutters.- All uncoated, unpainted, ferrous metal surfaces of each sealer, stretcher and cutter (see 3.3) shall be cleaned process C-1, thoroughly dried, and coated with type P-2 or P-6 preservative and packaged method I in accordance with MIL-P-116.

* 5.1.1.2 Seals.- Quantity of seals specified in tables I to VI, in one type and size only, shall be packaged in accordance with manufacturer's standard practice. The package shall then be sealed in a polyethylene bag constructed from minimum 0.001 inch thick polyethylene conforming to type I of L-P-378.

5.1.1.3 Strapping.- Not less than ninety pounds of type I, class B strapping shall be packaged in a coil in accordance with the level A requirements of QQ-S-781.

* 5.1.1.4 Operating instructions.- Each set of operating instructions shall be secured in location as shown on Drawings 5-7-64 through 5-7-66.

* 5.1.2 Level C.- Kit components shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A.

* 5.2.1 Level A.- Components of each kit and operating instructions (see 3.3), preserved and packaged as specified in 5.1, shall be placed and secured in the case in their proper location and the case closed and strapped in accordance with the applicable kit box drawing (see 3.3.3). Unless otherwise specified (see 6.2), no over-packing is required.

5.3 Marking.- In addition to any special marking required by the contract or order, each case shall be marked in accordance with MIL-STD-129, as indicated on kit box drawing.

6. NOTES

* 6.1 Intended use.- The strapping and sealing kits are used for securing and reinforcing bales, boxes, bundles, crates, and similar items, and for securing material within freight cars or trucks for shipment.

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* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type kit required (see 1.2).
(c) When preproduction sample is required (see 3.1).
(d) Selection of applicable levels of preservation and packaging (see 5.1).
(e) When over-packing is required (see 5.2.1).

6.3 Alternate component.- The term "or equal" shall be interpreted to mean that the alternate component furnished shall be dimensionally and functionally equal to, and of equal or better quality than the component referenced by the manufacturer's part number.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 3540-0075

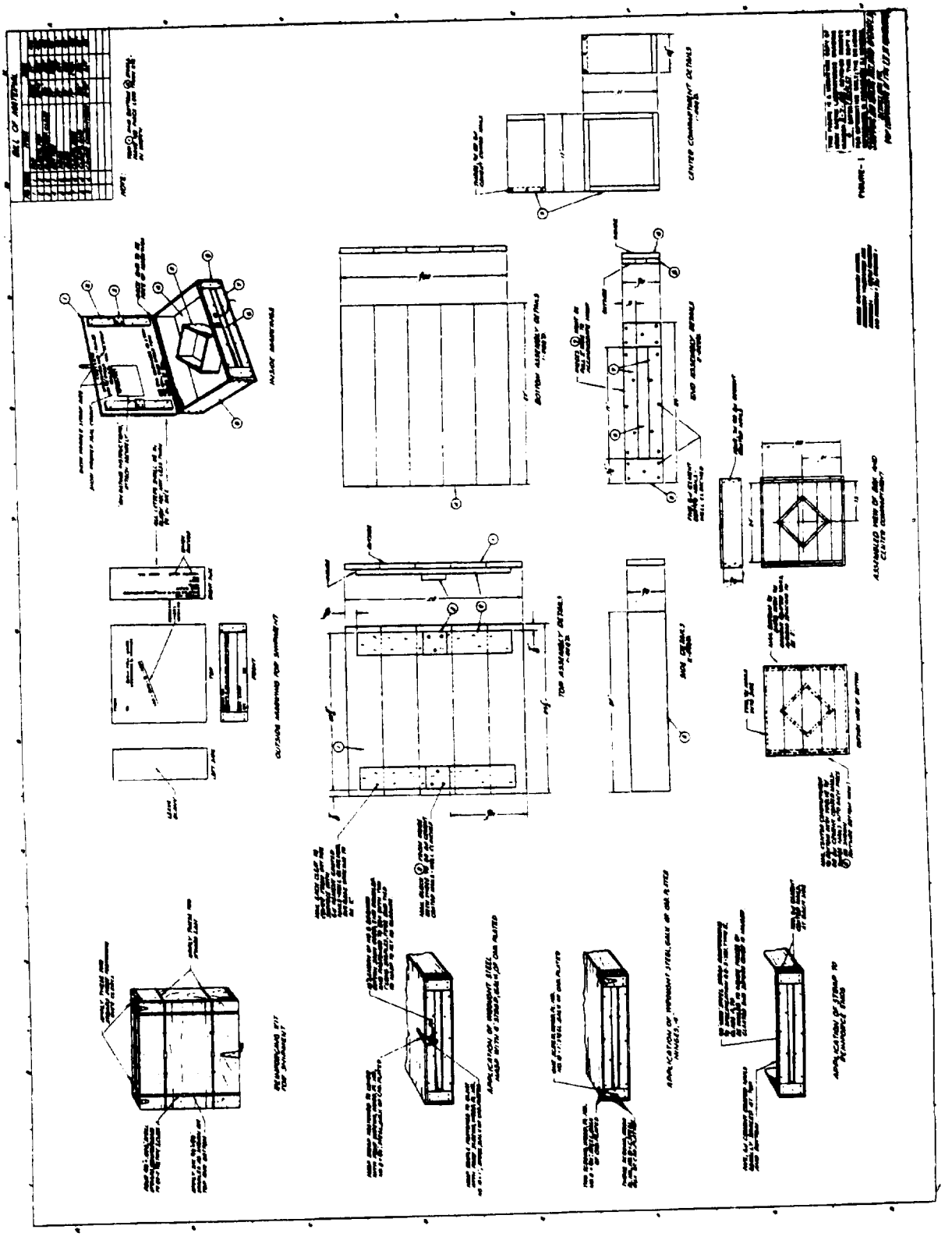
Review activities:

Army - MD, SM
Navy - SA
Air Force - 84, 85

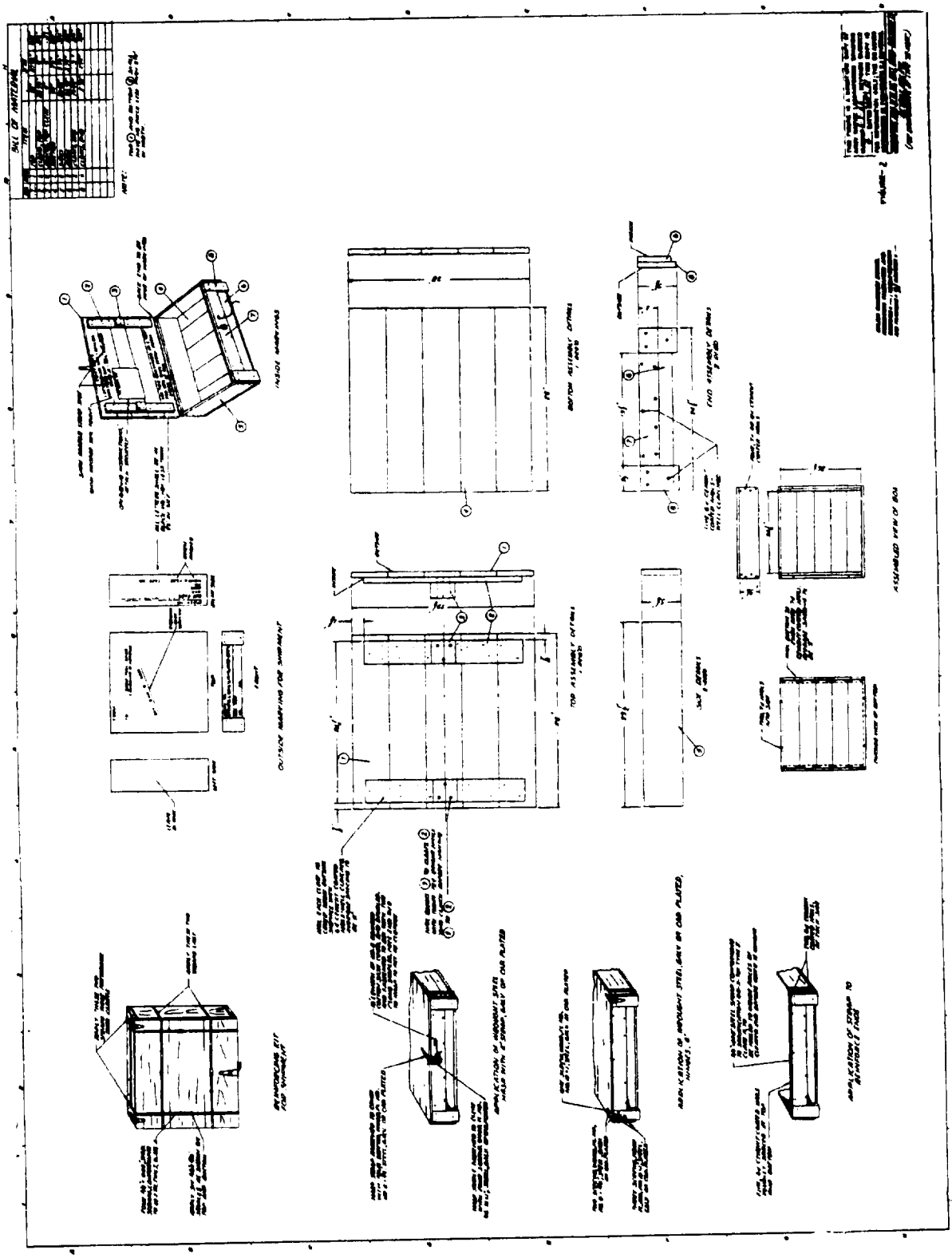
User activities:

Army - EL, ME
Navy - AS, MC, SH, YD

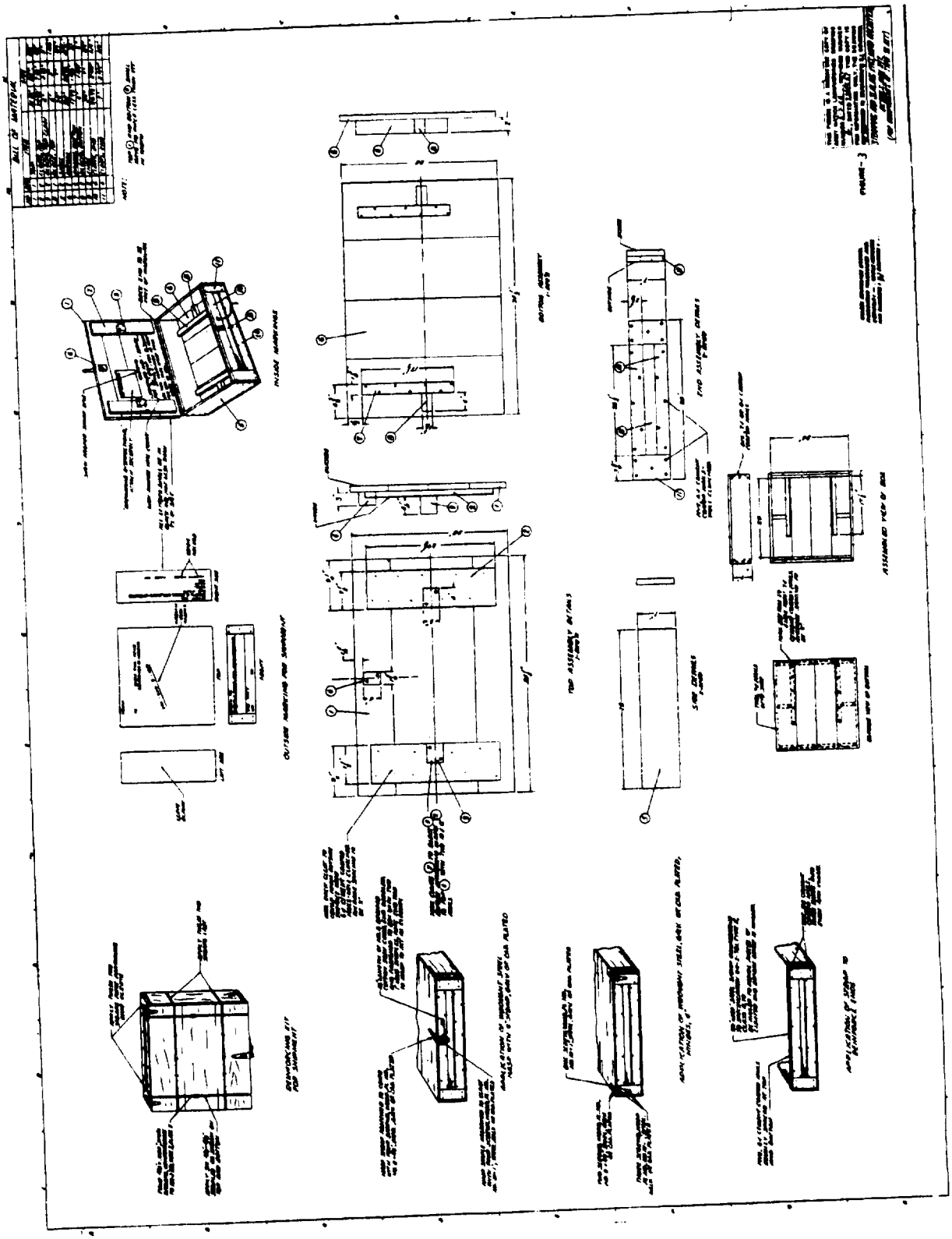
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Strapping and Sealing Kits, Hand-Operated MIL-S-43104B

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 OCT 64 1426

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